Prediction in Mixture-of-Experts based LLMs

Pierre-Louis Filoche

Supervised by Loris Marchal and Pablo Piantanida
ILLS Montreal, Canada
Master internship
May-July 2025



Motivation and Context

- LLMs keep growing in size → high inference cost
- MoE activates only a **subset of experts** per token during the forward pass
- Only uses $\frac{n}{N}$ of the model weights (n activated experts over N)
- Computational sparsity but not memory sparsity

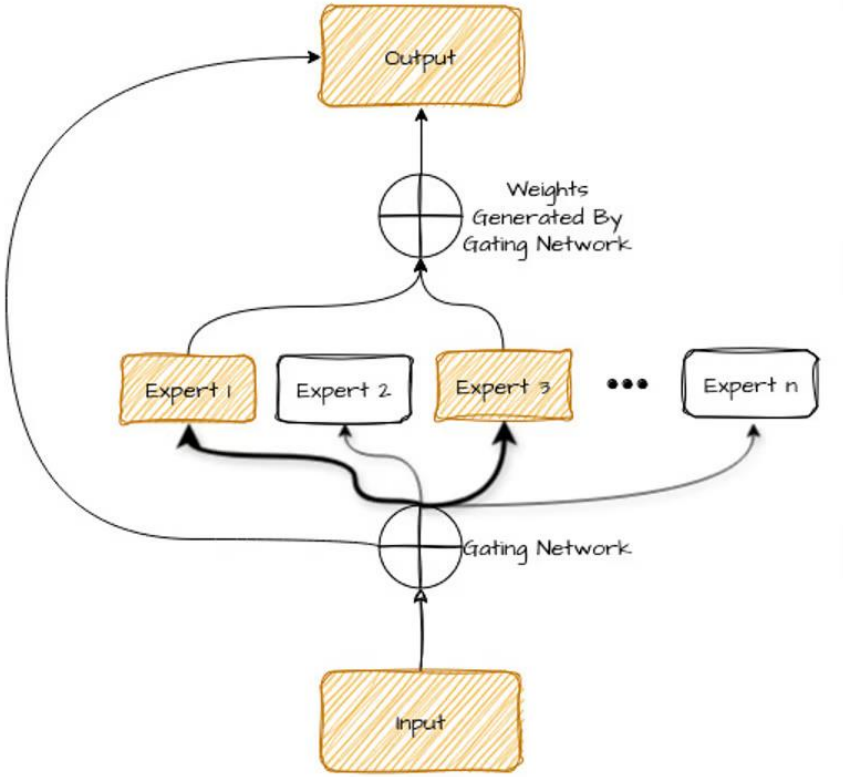
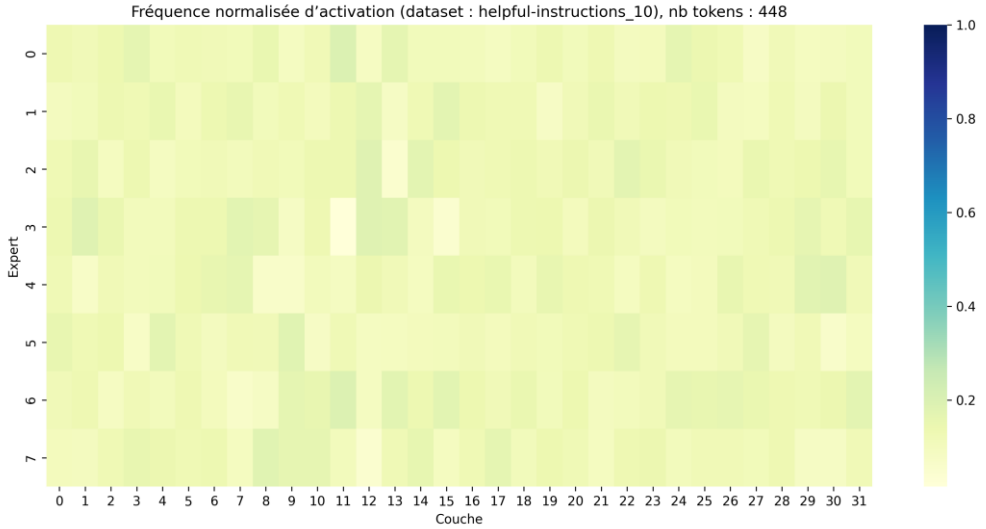


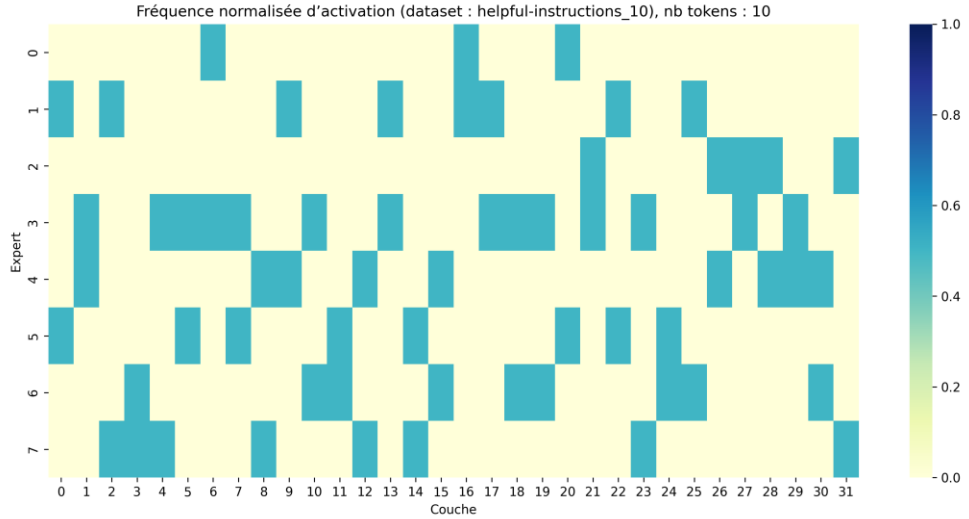
fig.1 – MoE Architecture

Mixtral 8x7b
n = 2
N= 8

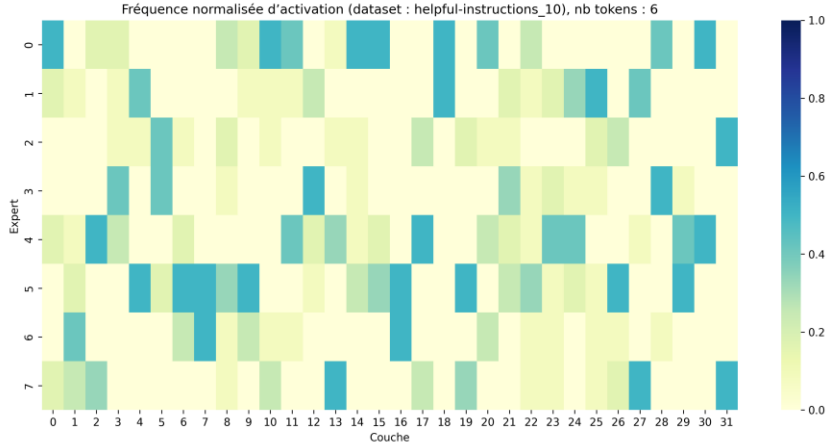
Motivation and Context



Mean over many tokens



Start token : determinist



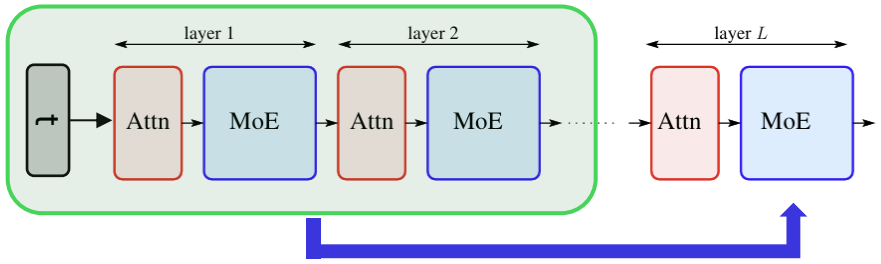
« , » token, complicated context

- Expert routing path depends on context
- Token _id might carry useful information

fig.2 - Expert use between tokens

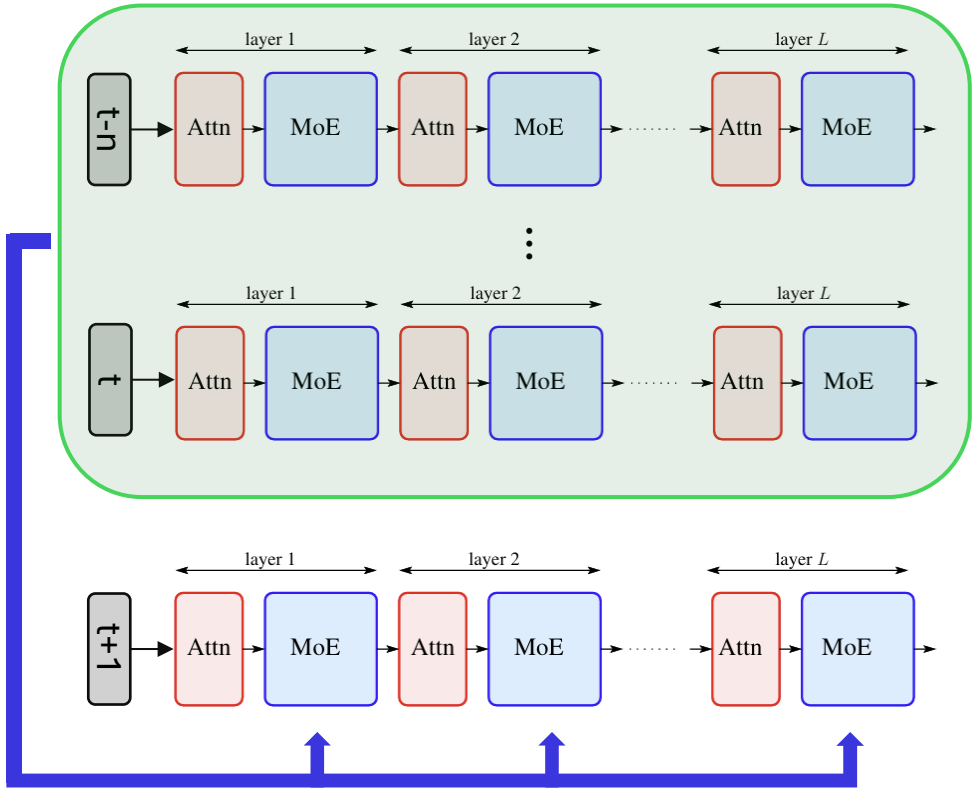
Objectives

Spatial prediction



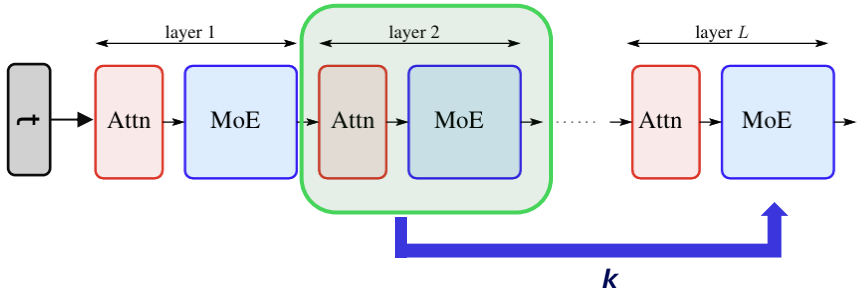
Predicting expert routing within a token, across layers

Temporal prediction



Predicting expert routing across tokens

Gating Network



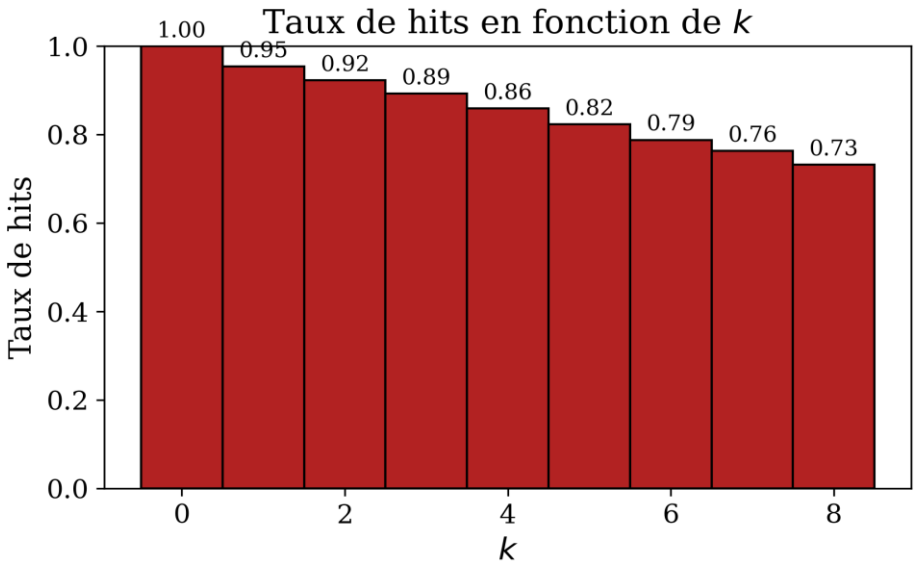


fig.3 – Prediction top-1

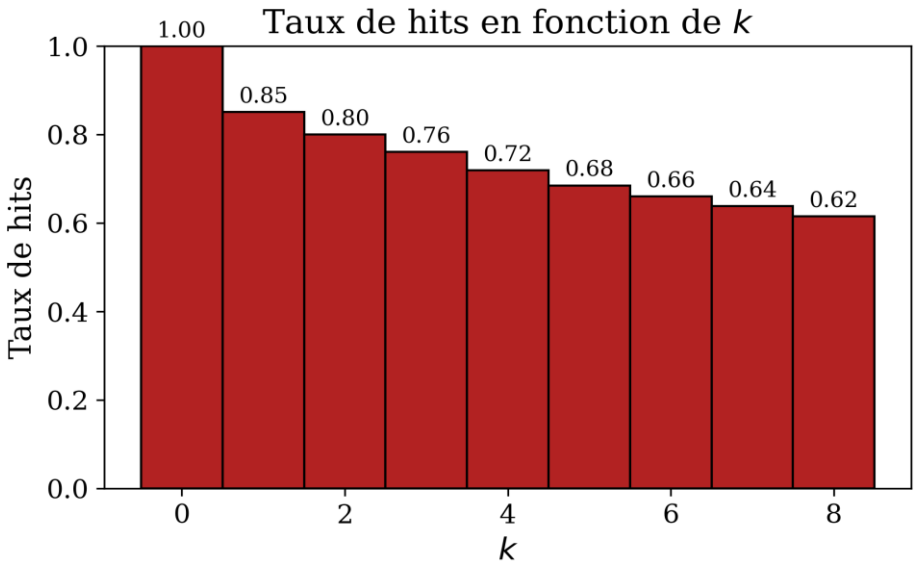
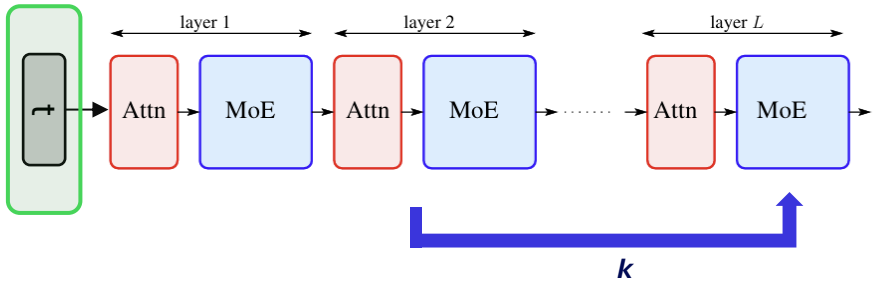


fig.4 – Prediction top-2

Statistical predictor



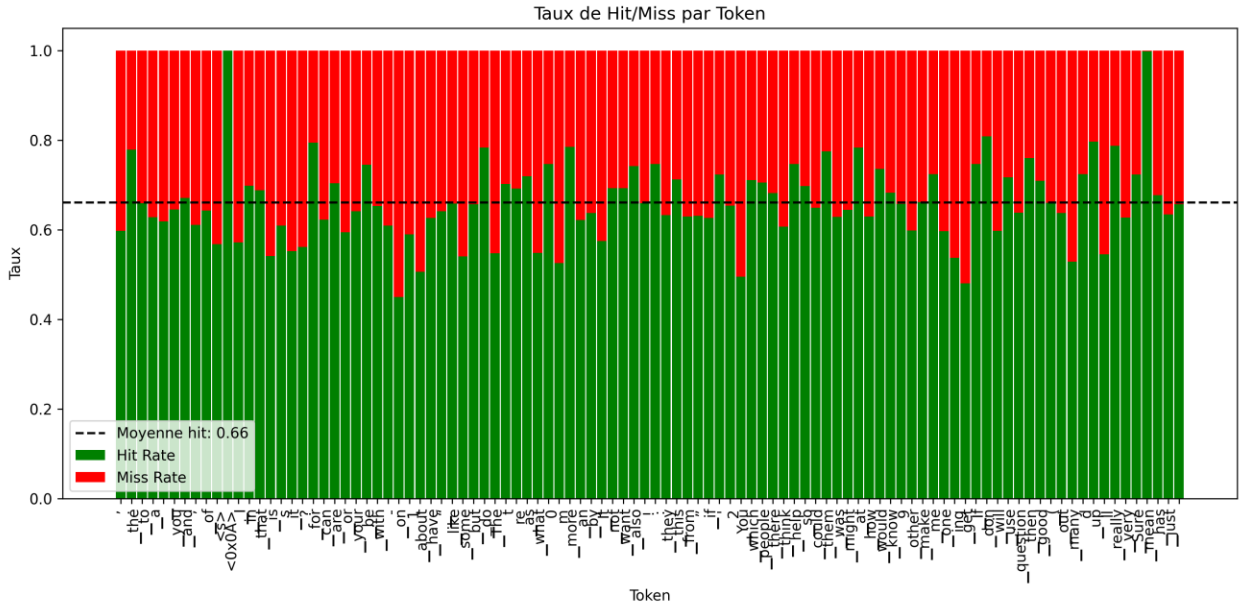
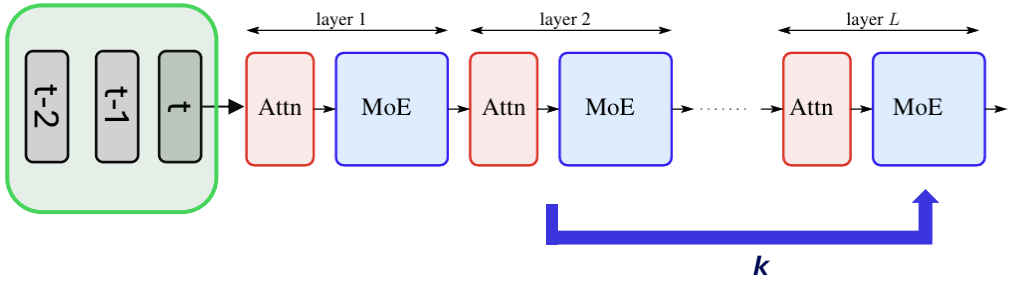


fig.5 – depth = 1

Statistical predictor



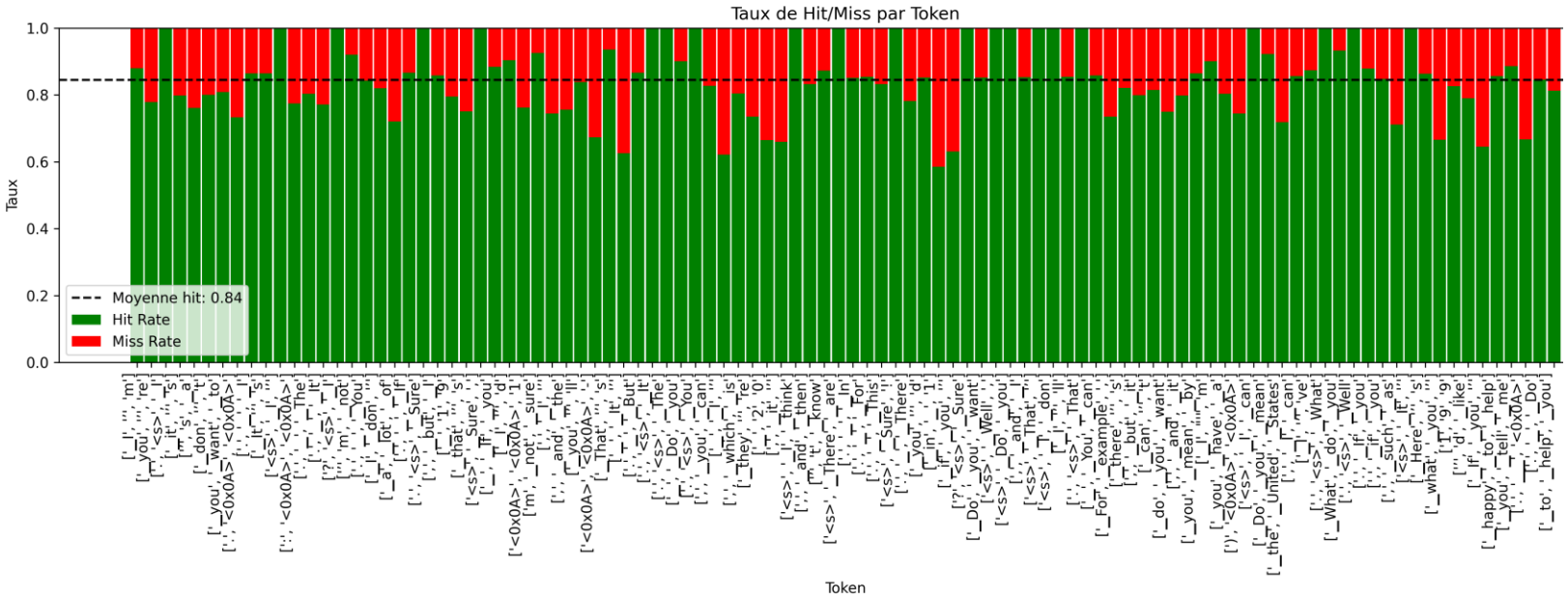
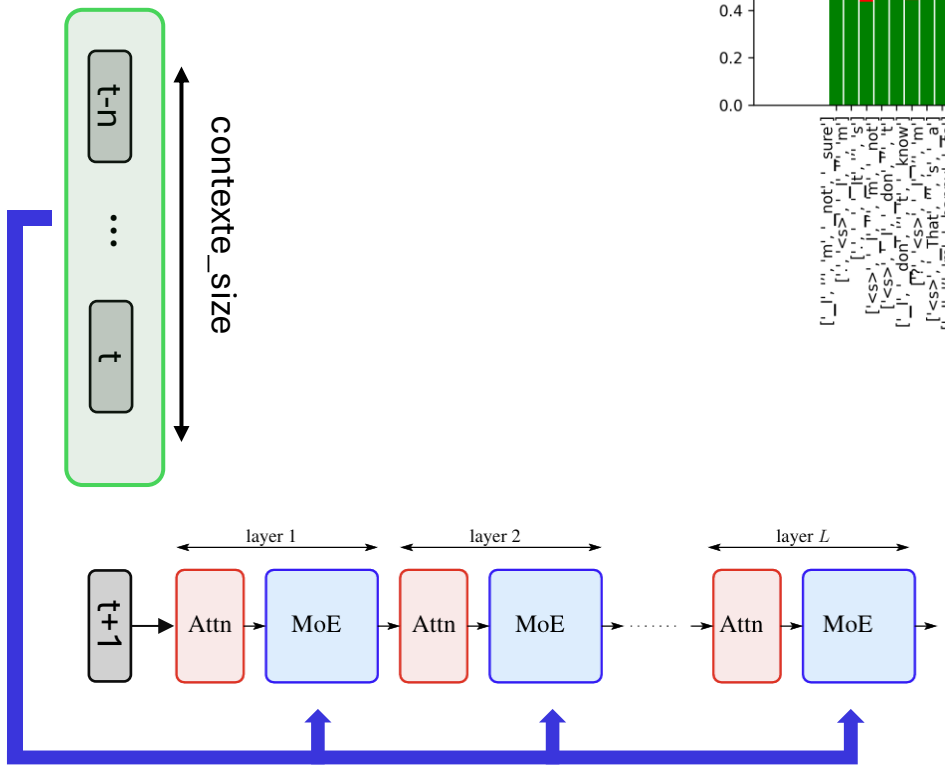
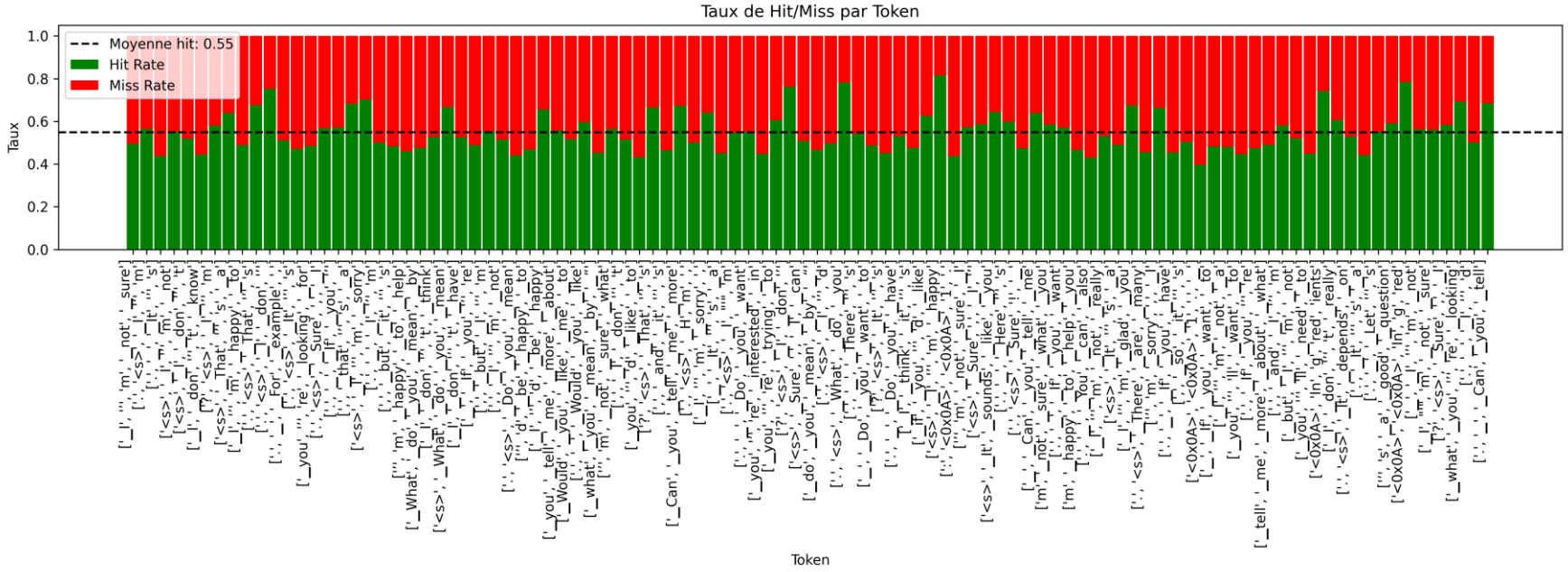


fig.6 – depth = 3

Statistical predictor





Down to **55%** for context_size = 5

Neural Predictors

MLP

Entrée aplatie

↓

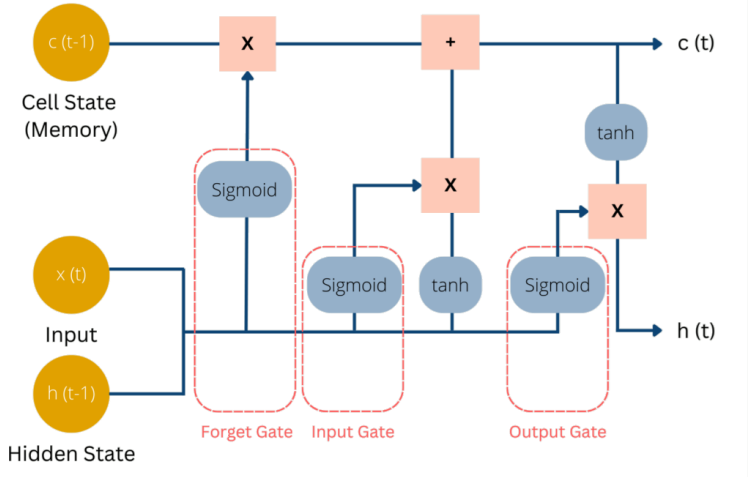
couches linéaires / ReLU

↓

logits experts

- **Limites :**
  - Concaténation brute (tokens + experts)
  - Croissance rapide de la taille d'entrée
- **Résultat :** ~51.6% top-1 (vs hasard 25%)

LSTM



- **Entrées :** embeddings réduits
- **Avantages :**
  - Exploite dépendances temporelles
  - Précision ↑ avec longueur du contexte
- **Résultat :**
  - Top-1 = **64.6%** Top-2 = **33.0%**

Conclusion

- The token ID and its context carry **useful information** about expert routing paths.
- **Statistical predictors** can exploit this information,
  but they **scale poorly** with vocabulary size and context length.

Perspective

Small Neural Network :

- Different prediction strategy
- Take advantage of expert patterns

Annexe

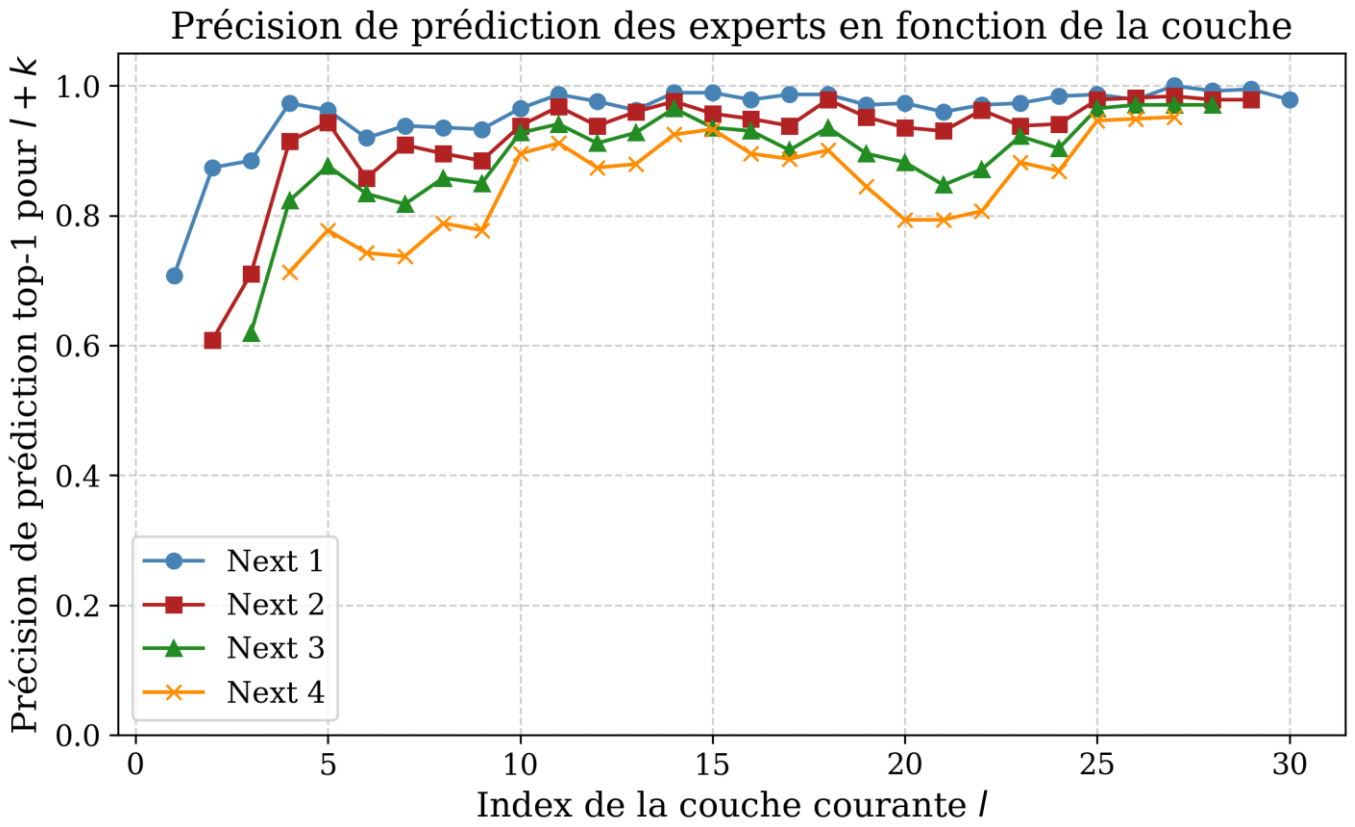


fig.A1 – Prediction top-1 gating-function

Annexe

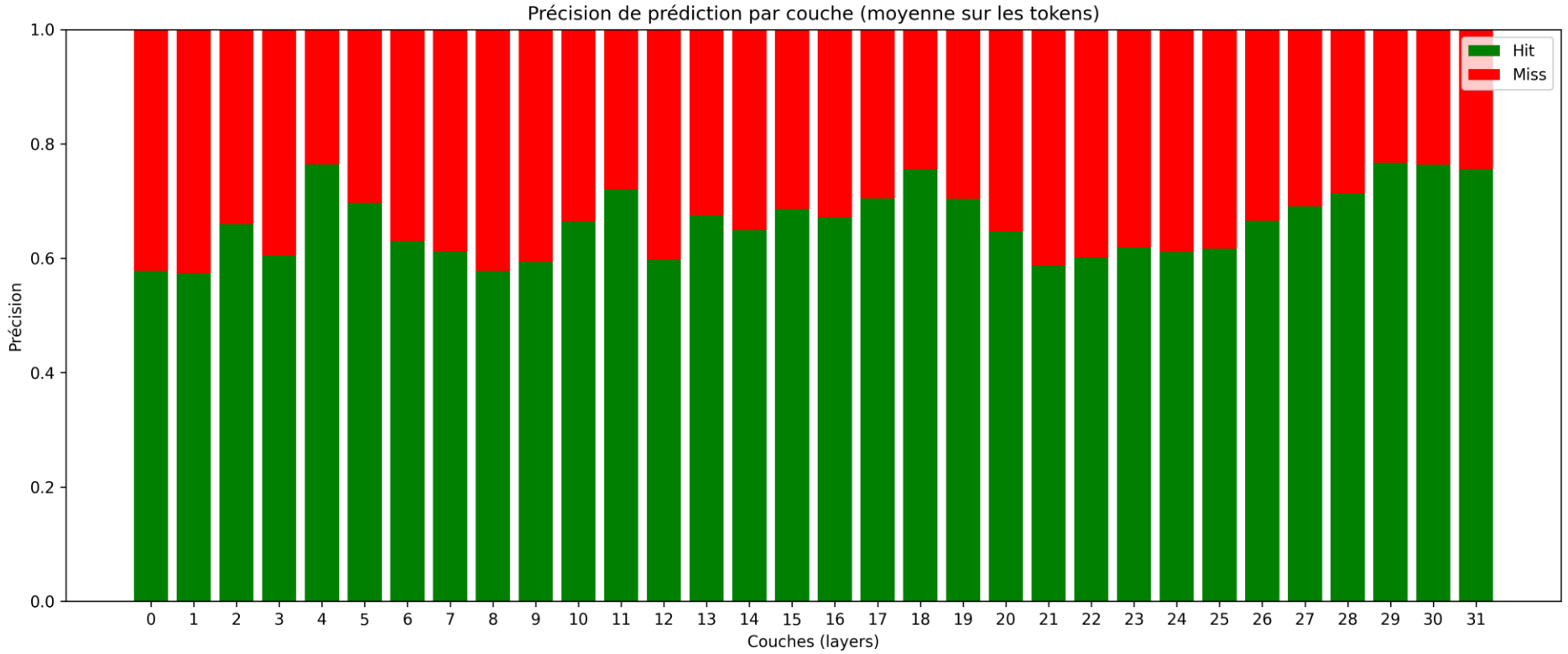


fig.A2 – Prediction top-2, prof=1

Annexe

Mixtral 8-7B model :

- 7B parameters / experts
- 12.9B active parameters per forward pass
- total of 45B parameters

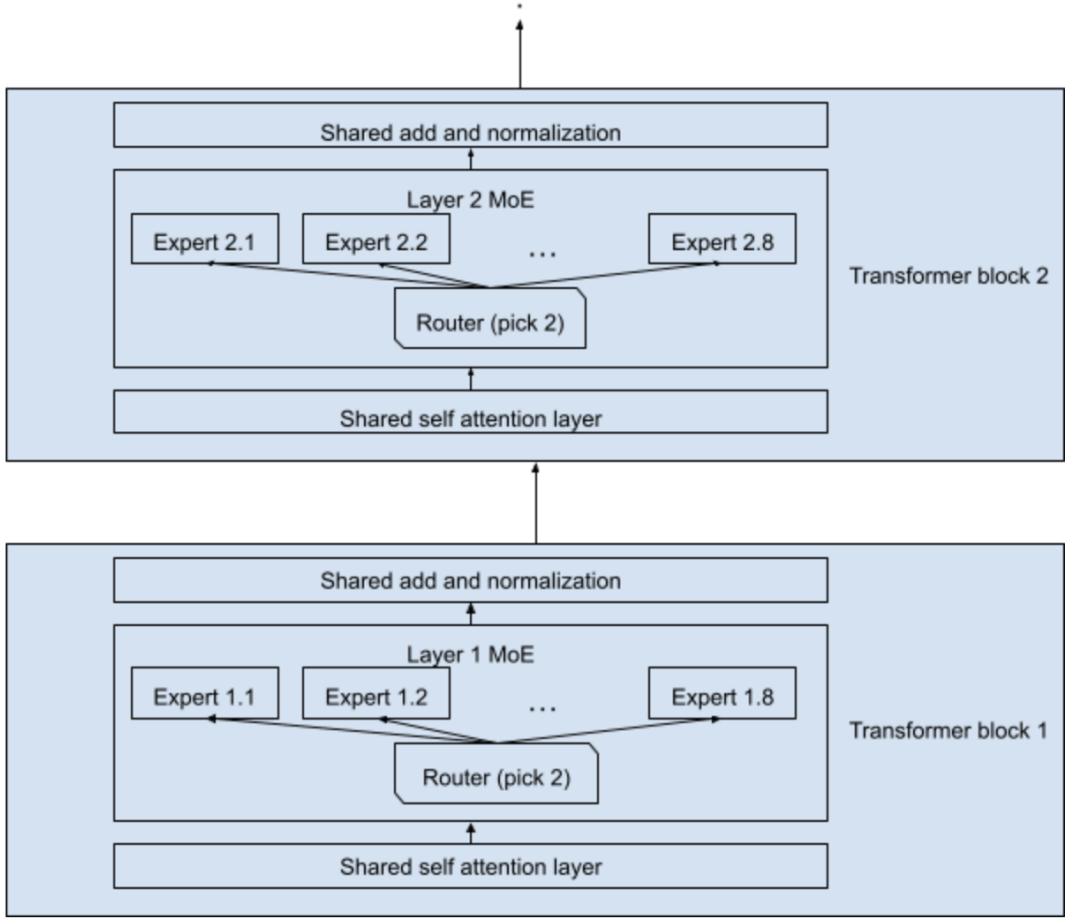


fig.A3 – Mixtral Architecture

Annexe

$$\text{Top-1} = \frac{1}{N} \sum_{\ell=1}^{N} \mathbf{1}\left[\hat{e}_\ell^{(1)} \in E_\ell\right],$$

$$\text{Top-2} = \frac{1}{2N} \sum_{\ell=1}^{N} \left(\mathbf{1}\left[\hat{e}_\ell^{(1)} \in E_\ell\right] + \mathbf{1}\left[\hat{e}_\ell^{(2)} \in E_\ell\right]\right).$$

fig.A4 – Metrics